

Design and Development of a Rear-Mounted Inductive Wireless Charging System for Electric

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Abstract:

Vehicles are everywhere now, and every- one wants charging to be easier, safer, and less of a hassle. The old-school wired chargers? They're clunky, you have to plug and unplug them yourself, and honestly, they wear out pretty fast. Plus, there's always that nagging worry about safety. That's why a lot of people are looking at wireless charging—just park your car and let electromagnetic induction do the work. But here's the catch: most wireless charging setups stick their coils right underneath the car. That sounds smart at first, but it actually creates two big headaches. First, the gap between the car and the ground messes with the magnetic connection, making everything less efficient. Second, the way those coils sit below means the magnetic field can heat up the metal parts of your car, which isn't just bad for performance—it's a fire risk.

So, this project set out to fix those problems. The main goal? Move the coils somewhere smarter and make the whole thing safer and more efficient. Instead of putting the coils under the car, we moved them to the back. Both the transmitter and receiver coils now sit at the rear, not the bottom. This tweak does a lot: it shrinks the air gap, makes lining up the coils easier, strengthens the magnetic link, and keeps that flux away from the car's metal frame

Keywords: Wireless Power Transfer; Electric Vehicles; Inductive Charging; Magnetic Coupling; Rear-Mounted Charging; Efficiency

1.INTRODUCTION

The high rate of electric cars (EVs) development is conditioned by the growing environmental issues, the decrease in the dependence on fossil fuels, and the necessity to reduce carbon emissions in the transport industry to a minimum [1]. The technology of EV has changed considerably during the last years, but there is a severe problem of efficient and safe charging infrastructure

. Traditional conductive (plug-in) charging solutions have the drawbacks of connector wear, safety concerns, inconvenience to users, and low accessibility [6], [7].

One of the alternative systems to the traditional wired charging systems has turned out to be Wireless Power Transfer (WPT) or Inductive Power Transfer (IPT) [4], [6]. WPT works on the law of electromagnetic induction introduced by Faraday in which power can be transferred inside a magnetic field without electrical contact between a transmitter coil and a receiver coil [5], [8]. This is a contactless mechanism that makes the use more convenient, minimizes mechanical wear, and contributes to the increased safety of operation.

Nevertheless, the performance of the wireless charging systems is very much determined by the alignment of the coils, the distance between the air gaps, the compensation topology with the magnetic coupling coefficient [6]. Research has indicated that the misalignment of transmitter and receiver coils contributes a great deal towards low efficiency and leakage of magnetic fields [4], [6]. Besides, during high power transfer the magnetic field emissions (MFE) can cause interaction with the metallic parts of the vehicle as well as with the adjacent structures and buildings that could become heated and cause safety issues [5]. Studies have shown that the intensity of the magnetic field and the misalignment condition have a direct effect on the efficiency of a given system and the amount of electromagnetic exposure [5]

The current systems adopt bottom-mounted coils mostly in the case of static wireless charge [4], [6]. This is on the one hand effective, but on the other, this arrangement suffers a loss of coupling as ground clearance increases, and as the arrangement becomes sensitive to alignment [6]. Moreover, exposures to electromagnetic fields should be in agreement with safety standards so as to prevent undue thermal and biological impacts [5]. Following these limitations identified, this paper presents a more refined coil architecture of the rear-mounted coil with a view of enhancing the magnetic coupling performance, limiting the adverse metal interaction, and eliminating thermal risk.

II. LITERATURE SURVEY

The technology of Wireless Power Transfer (WPT) has gained great research on an alternative to the traditional conductive charging systems used to charge electric vehicles (EVs). Some scholars have explored the coil configurations, compensation methods, efficiency minimization, safety issues and the magnetic field interactions.

To improve the efficiency of power transfers in dynamic EV charging systems, Mohamed et al. [1] suggested a wireless charging system with two receiver coils. Their article provided mathematical model to study the variation of mutual inductance and they showed that the charging power was improved on multiple coil configurations. Nevertheless, their research primarily concerned the problem of dynamical charging and was not concerned with the structural placement issues of vehicle-mounted coils.

Kumar et al. [4] provided an overview of the power transfer technologies in WPT that involved inductive power transfer (IPT), resonant inductive coupling, and different topologies of compensation series series series (SS) networks and LC networks. In the study, alignment of the coils, air gap distance and electromagnetic interference (EMI) have all been identified as important factors in terms of system efficiency. It has also brought out the fact that misalignment lowers coupling coefficient and enhances flux leakage

Rayan et al. [6] also talked of various coil designs and compensation methods of enhancing wireless charging. The paper has determined that the key contributors to poor performance of the system are coil misalignment and higher air gap. Minimization of the magnetic losses and the optimization of the coil geometry in favor of the coupling efficiency were also important aspects highlighted by the authors.

Under the research, McIntyre et al. [5] measured the magnetic field emissions (MFE) of vehicle-mounted WPT-based systems and examined the issues of safety concerning the electromagnetic exposure. Their experimental findings revealed that, misalignment enhances magnetic field emission and could affect thermal things. The paper has found that the safety standards should be significantly taken into consideration in high-power EV wireless charging systems.

Gaikwad et al. [8] and Mohan et al. [7] adopted a practical application of wireless charging systems using IPT and emphasized that electromagnetic induction of the coil placement and accuracy of alignment are very sensitive.

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Table 1: Key Design Parameters of Rear-Mounted Wireless EV Charging System

Parameter	Description	Value / Effect
Coil Placement	Position of transmitter and receiver coils	Rear-mounted (vehicle rear side)
Air Gap Distance	Distance between coils	Reduced compared to bottom-mounted system
Operating Frequency	Frequency of AC supply for induction	~14.22 Hz (low frequency for safety)
Coupling Coefficient (k)	Magnetic linkage between coils	Higher due to better alignment
Power Transfer Method	Type of wireless charging	Electromagnetic Induction
Compensation Network	Circuit used for resonance	LC (LNC) Resonant Circuit
Eddy Current Loss	Loss due to metal interaction	Reduced in rear-mounted system
Heating Effect	Heat generated due to magnetic interaction	Minimized
Alignment	Ease of positioning coils	Easier and controllable
Efficiency	Overall power transfer efficiency	Improved compared to conventional system

III. System Architecture

The proposed system will have two parts, the transmitter part which will be at the charging station and the receiver part at the rear part of the electric vehicle. The system is also electro-magnetically inducted where electrical power is sent wirelessly by a magnetic field created between two coils.

The transmitter unit is found in the charging station or charging dock fitted at the rear side of the car. It is the section that generates the alternating magnetic field necessary in the transfer of wireless energy

A DC power source that is provided by the electrical grid starts the system. It is this strength that provides the main energy source to the wireless charging system.

The AC input power is first changed into DC by use of an AC-DC rectifier circuit. A filter circuit is employed to filter off voltage ripples and offer a smooth DC signal to the next stage.

High-frequency inverter converts the DC voltage which goes through the filter into high-frequency AC. This step is required

since induction of electromagnetism needs alternated current to produce a varying magnetic field.

The compensated network venerates the high-frequency AC signal to the transmitter resonant coil. The compensation network is a resonant circuit comprising of both inductors and capacitors which enhance efficiency in energy transfer and minimizes loss of reactive power. The transmitter coil (L1) induces alternating magnetic field that travels through the air gap to the receiver coil that is placed on the vehicle.

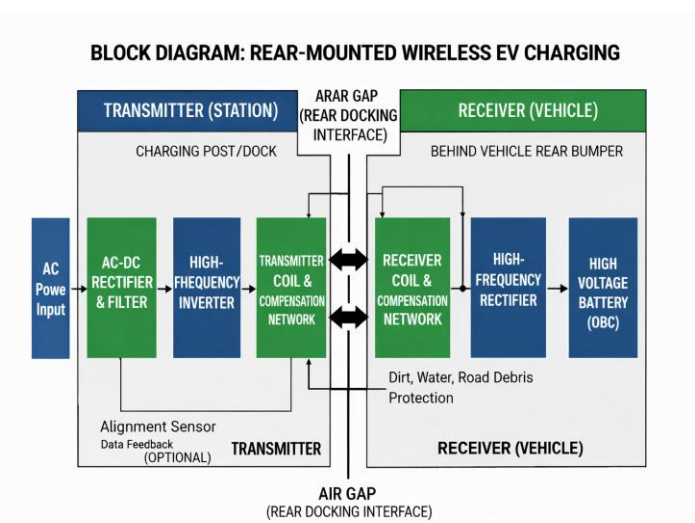


Figure 1: Block Diagram of Rear-mounted wireless EV charging system

Table 2: Major Components of the Proposed System

Component	Function	Role in System
AC Power Supply	Provides electrical energy from the grid	Acts as the primary energy source for the system
AC-DC Rectifier & Filter	Converts AC to DC and removes ripples	Supplies stable DC power to the inverter stage
High-Frequency Inverter	Converts DC into high-frequency AC	Generates alternating current
Transmitter Coil (L1)	Produces alternating magnetic field	Transfers energy wirelessly
Compensation Network (LC)	Maintains resonance condition	Improves power transfer efficiency
Air Gap (Rear Interface)	Space between coils	Enables wireless energy transfer
EV Battery Pack	Stores electrical energy	Supplies power

On-Board Charger	Manages charging process	Controls safe
DC-DC Converter	Regulates voltage and current	Ensures suitable charging conditions for the battery

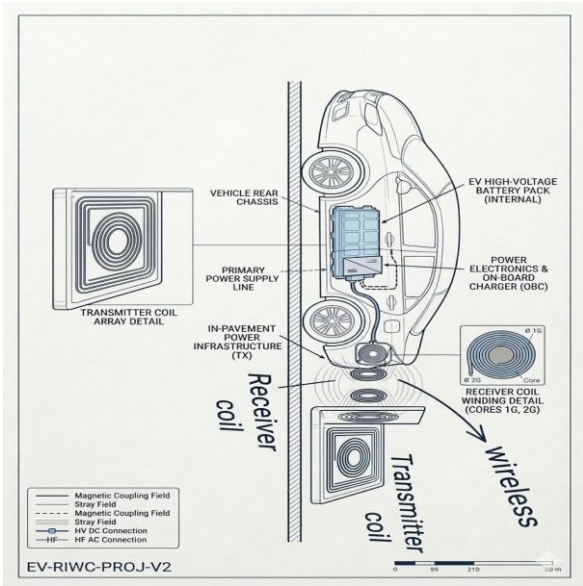


Figure 2: Rear-mounted inductive wireless charging system architecture for electric vehicles.

The figure 2 depicts the inductive wireless charging system that is mounted on the rear side of the electric vehicles. The device is comprised of two major parts: a transmitter coil that is placed at the charging station and a receiver coil that is placed to the back part of the electric vehicle. The in-pavement power infrastructure supplies electrical energy to the transmitter coil using electrical energy produced by the main power supply. The alternating current passing through the transmitter coil will produce an alternating magnetic field

IV. METHODOLOGY

This formula forms the law of electromagnetic induction. It demonstrates that a voltage occurs in the receiver coil with change in magnetic flux. In our suggested system, the transmitter coil produces a changing magnetic field and this causes the rear-mounted coil (receiver) to produce voltage.

$$E = -N \frac{d\Phi}{dt}$$

Where:

- **E** = Induced electromotive force (Voltage) (Volts)
- **N** = Number of turns in the coil
- **Φ (Phi)** = Magnetic flux (Weber, Wb)
- **t** = Time (seconds)

Magnetic flux is the measure of the quantitative level of magnetic field through the coil field. In the suggested design, the more the air gap between the coils is reduced, the stronger the magnetic field is generated and the stronger the flux is, so the better energy transfer.

$$\Phi = B \cdot A$$

Where:

- Φ (Φ) = Magnetic flux (Weber, Wb)
- B = Magnetic field strength (Tesla, T)
- A = Area of the coil (m^2)

This is the mutual inductance between coils of transmitters and receivers. It is dependent on the coupling coefficient and inductance of the two coils. The coupling coefficient is enhanced through improving the alignment in the rear-mounted system that enhances mutual inductance and transfer of power.

$$M = k \sqrt{L_1 L_2}$$

Where:

- M = Mutual inductance (Henry, H)
- k = Coupling coefficient (0 to 1)
- L_1 = Inductance of transmitter coil (H)
- L_2 = Inductance of receiver coil (H)

This formula is used to calculate the resonant frequency of the LC compensation network. Working at resonance enables the transfer of power between the transmitter and receiver coils to be the most efficient since reactive losses are minimized and the system becomes more efficient.

$$f = \frac{1}{2\pi\sqrt{LC}}$$

Where:

- f = Resonant frequency (Hertz, Hz)
- L = Inductance (Henry, H)
- C = Capacitance (Farad, F)
- π = Constant (≈ 3.14)

This formula is an indicator of efficiency of the wireless charging system. It is the proportion of power that is received by the battery to the input power given. The proposed rear-mounted system is more efficient, since it is better coupled and the losses are minimized.

$$\eta = \frac{P_{out}}{P_{in}} \times 100$$

V. RESULTS AND DISCUSSION

The magnetic field then crosses through the air between the transmitter and the receiver coils and causes voltage in the receiver coil by the principle of electromagnetic induction. The receiver coil is mounted vertically behind the rear bumper of the vehicle to enhance the alignment of the coils, and also to decrease the distance between the coils. The generated electrical power is then passed on to the power electronics and onboard charger (OBC) in the vehicle.

The onboard charging system also controls the induced power and delivers it to the high-voltage battery pack that stores the electrical energy to run the vehicle. The magnetically exerted field of coupling and stray magnetic field during the transfer of wireless power is also represented in the diagram. When the coils of the transmitter and receiver are properly aligned, the coupling coefficient can be maximized, power transfer efficiency can be maximized, and loss of energy can be reduced.

The system minimizes eddy current losses and heat generation by mounting the receiver coil at the rear of the vehicle rather than the bottom thereby minimizing vehicles interference with metallic parts of the vehicle chassis. This arrangement is also safer and more effective in charging wireless electric vehicles charging systems.

Fig 2 Rear-mounted wireless EV charging system Moreover, distance of the transmitter and receiver coils, alignment of the coils, and coupling coefficient of coils are the key determinants of the efficiency of the wireless charging system. The improved power transfer efficiency is achieved when the coils are aligned well, and it leads to a higher magnetic flux linkage.

The receiver coil is placed vertically at the rear side of the vehicle in the proposed system and this permits the vehicle to be brought close to the charging station.

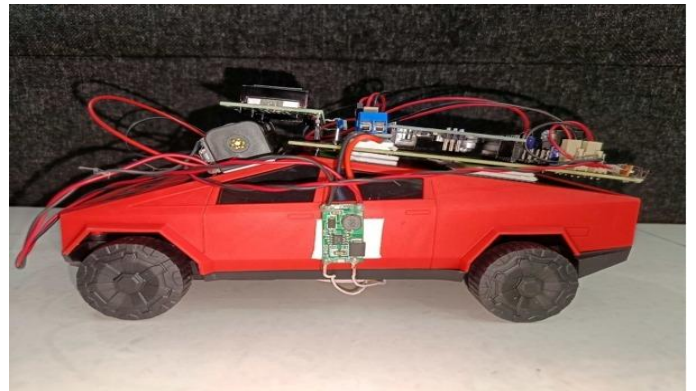


Fig 3 Prototype model of the rear-mounted wireless EV charging system.

The rear-mounted wireless EV charging system proposed was tested and confirmed with the help of a prototype. The system will be made up of a transmitter coil, which will be located at the charging station, and a receiver coil, which is on the back of the vehicle. Alternating current is fed to the transmitter coil so as to induce a time-varying magnetic field. Once the vehicle is in close proximity to the station the magnetic flux is captured by the receiver coil which produces an alternating voltage as a result of

electromagnetic induction. A rectifier is used to transform this induced voltage into DC and a DC-DC converter is used to regulate the voltage to charge a battery. The hardware outcomes demonstrate that a good alignment of the coils enhances the coefficient of the coupling and increases the efficiency of the transfer of power. The coil is mounted on the rear thus eliminating the air gap between coils resulting in improved performance. It also reduces the contact with metallic components and therefore lowers the eddy current loss and heating effects. Under congruent conditions, the system proves to be stable and efficient in terms of power transfer. The prototype also proves that the proposed design is safe and reliable. Moreover, the positioning at the back makes the alignment of the charging easier. In general, the hardware verification is used to demonstrate the viability and efficiency of the rear-mounted wireless EV charging system.

Table1: Comparison between Bottom-Mounted and Rear-Mounted Wireless EV Charging Systems

Bottom-Mounted Wireless Charging	Rear-Mounted Wireless Charging (Proposed)
Receiver coil is placed under the vehicle.	Receiver coil is placed at the rear side of the vehicle.
Large air gap due to vehicle ground clearance.	Smaller air gap between transmitter and receiver coils
Coil alignment is difficult.	Coil alignment is easier.
Magnetic coupling efficiency is lower.	Magnetic coupling efficiency is higher
Magnetic field interacts with metal chassis.	Less interaction with metallic parts.
Higher eddy current losses.	Reduced eddy current losses.
Higher heating risk.	Heating effect is reduced.
Lower power transfer efficiency.	Improved power transfer efficiency.
Difficult maintenance under vehicle.	Easier maintenance at rear side.
Safety risk is comparatively higher.	System safety is improved.
Dirt, water, and road debris may affect the coil under the vehicle.	Rear-mounted coil is less exposed to road debris.

The rear-mounted wireless charging system proposed was tested to assess its functionality in regards to the coil positioning, magnetic interaction efficiency as well as heating. The findings show that the positioning of the receiver coil behind the vehicle enhances the positioning of transmitter coil and receiver coil better than the traditional bottom-mounted charging systems.

In the offered arrangement, the air distance between the coil of transmitters used and that of receivers can be minimized since the vehicle can be brought nearer to the charging station. The smaller the air gap, the higher the magnetic coupling coefficient and hence the more efficient the wireless charging system is in terms of the power transfer. The better coupling also leads to increase in the induced voltage in the receiver coil.

The other noteworthy observation is that eddy current losses are minimized. In wireless charging systems mounted on the bottom part of the car, the magnetic field can be in contact with metal parts that are present beneath the car, and therefore unwanted heating can occur. The coil of receiver is located on the back side of the car so that it communicates with the metallic chassis components as little as possible. Therefore, the suggested system demonstrates reduced heat production and better safety of operations.

The rear mount position is also easier to align the charging station to the vehicle. This ensures that the charging is convenient and reliable to the users. All in all, the offered Wireless charging system mounted on the rear is more efficient, less thermally sensitive, and safer than traditional wireless charging systems mounted on the bottom.

VI. CONCLUSION:

This paper introduces the design and the idea of an inductive wireless charging system mounted at the rear of the electric vehicles. The system proposed features the location of receiver coil at the backside of the vehicle, rather than beneath the chassis. This is done to minimize the air-gap between the coil of the transmitter and the receiver and enhance efficiency of the magnetic coupling. Consequently, this makes the transfer of power wirelessly more effective than the traditional systems that were mounted on the bottom.

In the study, the rear-mounted system is also found to minimize the contact of the magnetic field with the metal structures found beneath the vehicle. This assists in the reduction of eddy current loss, and also to minimize the unwanted heat generation and thus enhances the safety of the charging system.

Moreover, the receiver coil is placed at the back of the vehicle, and thus alignment of the vehicle to the charging station is easier which increases convenience of the user in charging.

In general, the developed rear-mounted wireless EV charging system is proven to be more efficient, less heating, and more operational safe. As such, the given setup is a potentially promising option of future wireless charging infrastructure in electric vehicles.

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